Lab1 - Transients in RLC Networks

1. Introduction

RLC networks are commonly used as bandpass and notch filters to selectively remove certain frequencies from the signal delivered to a load, in oscillators, and in circuits that used in energy storage and power conversion such as pulse discharge circuits and voltage multipliers. There are three different transient responses in RLC circuits – overdamped, critically damped, and underdamped. Engineers select the type of damping when designing circuits to achieve a certain performance.

In the case of the pulse discharge circuit such as some circuits used to send short, high voltage pulses to the lamp in a camera flash, a response is preferred that causes all of the energy stored in the inductor or capacitor to be rapidly dissipated by the resistor in RLC network. The capacitor in a series RLC network and the inductor in the parallel RLC network determine the amount of stored energy, which will determine the brightness of the camera flash. Thus, the objective in a pulse discharge circuit is to design an RLC network that releases most of the stored energy through the resistor within a specified amount of time, usually as short as possible. Thus, an engineer must consider both the natural frequency ω_o and the damping factor α when selecting components.

The transition between initial and final conditions for component voltages and currents is fastest in a critically damped circuit. You can always design a critically damped circuit, but to have an RLC network that operate always at the point where $\omega_o = \alpha$, requires components that (a) have extremely tight tolerances or are tunable (e.g., a trim potentiometer and variable capacitor) so that the magnitudes of the components can be tweaked to the required values, (b) that have very little dependence on environmental variables such as ambient temperature and (c) that behave as ideal component over the current, voltage, and frequency ranges of operation for the life of the circuit. As there are parasitic resistances, capacitances, and inductances in a number of components – even a resistor such as any of the ones from the Electronics Store should be modeled as an inductor in series with the parallel combination of a capacitor and the resistor to predict its operation over a reasonably wide range of frequencies. Hence, an RLC network that operates as a critically damped circuit is difficult to achieve.

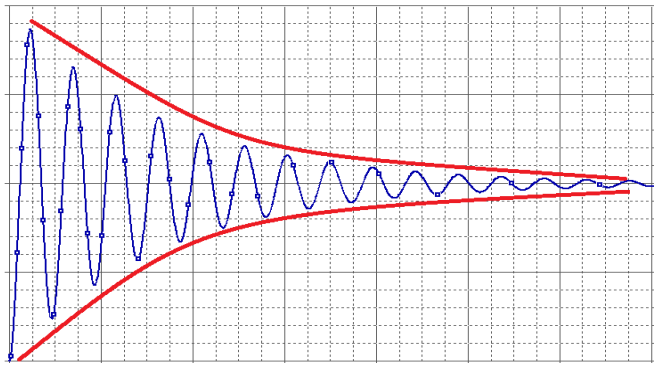


Fig. 1. Example of the response of an underdamped RLC network.

Although essential in certain oscillator circuits, the underdamped response can be undesirable as it can take a long time for the signal to reach steady-state. The signal in an underdamped RLC network will overshoot – the signal will temporarily change sign as the signal decays to zero in the case of a natural response or it can reach a much larger magnitude than the magnitude of the final steady-state signal in the case of a forced response. The overshoot, which is also known as ringing, oscillates at the damped frequency of the RLC network $\omega_d = \sqrt{\omega_o^{\,2} - \alpha^2}$ and the rate of the exponential decay of the oscillations is determined by the damping factor. The magnitude of the overshoot must be taken into consideration when determining the maximum voltage and current specifications for the components selected for use in the circuit. In addition to stressing components electrically, the overshoot of an underdamped circuit can also stress components mechanically because of rapid thermal expansion caused by a spike in the component's temperature as a result of the dissipated power. The stress can damage components, enhancing their non-ideal properties, and can reduce the components' useable lifetime.

In this experiment, we will explore how the value of α affects the ringing that occurs in an underdamped series RLC network. The circuit that will be analyzed is shown in Figure 2. α will be altered by varying the value of R1. Note that this circuit is a bit more complicated than a simple series RLC circuit. First, there is a Thévenin equivalent resistor R_{th}, which is in series with the voltage source V_1. This resistor is used to model the resistance of the arbitrary waveform generator, which is a non-ideal voltage source. Next, there is a resistor R_2 that is in parallel with the RLC network. This resistor is used to insure that the energy stored in the capacitor does not discharge directly into the arbitrary waveform generator, which could cause the current to flow into the voltage source. Depending on the voltage source used, this may cause the voltage source to turn off and could possibly damage the arbitrary waveform generator. Instead, the transient current generated while the capacitor discharges will be added to a dc current $V_1/(R_{th} + R_2)$. The values that will be used in this experiment for V_1, the inductor, the capacitor, and the resistors have been selected to insure that the arbitrary function generator will always act as a power generating element in the circuit. Lastly, a lumped component model for a non-ideal inductor is used – a series combination of an ideal inductor L_1 and a resistor $R_{parasitic}$, which is the resistance of the wire used to create the inductor.

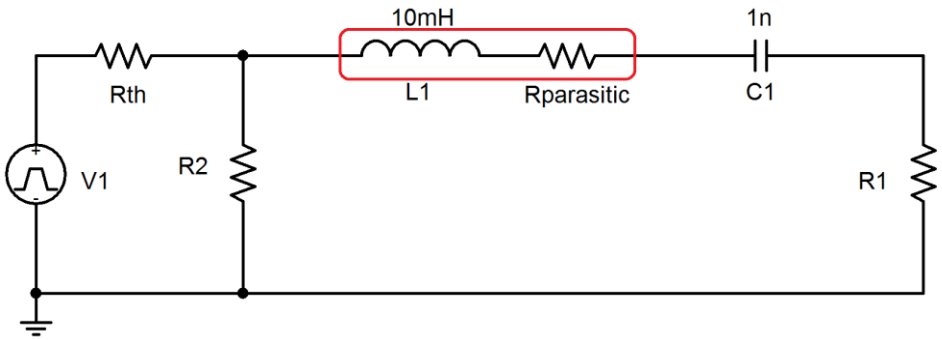


Fig. 2: Schematic of a series RLC circuit.

NOTE: Caution should be used when designing a RLC network to insure that the voltage and current specifications for the voltage source are not exceeded or it may be damaged.

2. Analysis – Step1

1. Write the transfer function for the voltage across C_1 versus the input voltage V_1.
2. Determine the natural frequency, damping factor, and damping frequency of the circuit shown in Figure 2 where $R_{th} = 50\Omega$, $R_2 = 47\Omega$, $R_{parasitic} = 100\Omega$ and a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$. Note that you should combine R_{th}, $R_{parasitic}$, R_1, and R_2 into one equivalent resistor before performing these calculations.
3. For the two cases a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$, determine the values for A_1 and A_2 in the equation for a forced response, $v_C(t) = [A_1\cos(\omega_d t) + A_2\sin(\omega_d t)]e^{-\alpha t} + Vs$ when $V_1 = 2V[1 + 2u(t)]$, where $u(t)$ is a unit step function. Note that $Vs \neq V_1$. The solution will be the same as the case where $V_1 = 4Vu(t)$ because the transient response is only produced by the step function, not the d.c. component of V_1.
4. Using MATLAB, plot the equations for the voltage across the capacitor developed in Step 3 on the same graph from t = 0 s to t = 0.25 ms.

3. Modeling – Step2

1. In PSpice, simulate the transient response of the circuit shown in Figure 2 where $R_{th} = 50\Omega$, $R_2 = 47\Omega$, $R_{parasitic} = 100\Omega$ when a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$. Set the amplitude of the voltage source V_1 such that it is a square wave with an amplitude that varies from 2 V to 6 V. The pulse width should be 0.25 ms and the period should be 0.5 ms.
2. Graph the voltage across C_1, the voltage across R_1, and the voltage across R_2 for one period of the square wave. Either scale the curves so that the magnitude and shape of each signal is readable or add plots to obtain a stack of graphs of the signals.
3. Graph the current flowing out of V_1 to show that the voltage source is always a power generating element in the circuit.
4. Measure the period of the underdamped response of the voltage across the capacitor when a) $R_1 = 300\Omega$ and b) $R_1 = 3k\Omega$. Calculate the damped frequency ω_d using the period measured. Determine the percent difference between the expected value of ω_d calculated in Step 1 and the damped frequency calculated from the data graphed in PSpice.

4. Measurements- Step3

1. Measure the parasitic resistance $R_{parasitic}$ of a 10 mH inductor.
2. Construct a modification of the circuit shown in Figure 2. The modifications are as follows:

- Do **not** include R_{th} as the arbitrary function generator has a Thévenin equivalent resistance of 50 Ω.
- Do not include a 100 Ω resistor for $R_{parasitic}$ as this resistor is part of your (non-ideal) inductor.

3. Select the square wave with a 50% duty cycle on the arbitrary waveform generator. Set the frequency so that the period of the square wave is 0.5 ms. The amplitude of the voltage source should be set such that the output of V_1 oscillates between 2 V – 6 V.
4. Measure the voltage across C_1 and the voltage across R_1 when $R_1 = 300\Omega$ for one period. Determine the damped frequency ω_d of the transient response when the input voltage increases from 2 V to 6 V and, again, when the input voltage decreases from 6 V to 2 V. Calculate the percent difference between the damped frequency found in Step 1 and the measured frequency.
5. Measure the voltage across C_1 and the voltage across R_1 when $R_1 = 3k\Omega.$ for one period. Determine the damped frequency ω_d of the transient response when the input voltage increases from 2 V to 6 V and, again, when the input voltage decreases from 6 V to 2 V. Calculate the percent difference between the damped frequency found in Step 1 and the measured frequency.

5. Conclusion

1. To reduce the length of time that ringing of the voltage across R_1 can be observed:
   a. Should α increase or decrease?
   b. If R_1 is held constant, should the value of L_1 increase or decrease to reduce the transient response?
   c. Reconsider your answer in part b when a real inductor is used when R_1 is small (i.e., 300 Ω).
2. Given the relationships between energy, power, voltage, and resistance, do you expect that the energy stored in the capacitor will dissipate faster as R_1 increases? Why or why not?
3. Given the relationship between energy, charge, voltage, and capacitance, would you expect that the amplitude of the voltage overshoot in the transient response to increase or decrease if the capacitance of C_1 is increased while the magnitudes of the other components in the circuit are held constant?
4. If the d.c. offset is set to 0 V, why would it be inappropriate to use an electrolytic capacitor for C_1 ?